

Hanjun Park

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EDUCATION

Ph.D., Industrial and Systems Engineering Virginia Tech, Blacksburg, VA	08/2025
Graduate Certificate in Data Analytics Virginia Tech, Blacksburg, VA	
M.S., Industrial Engineering Seoul National University, Seoul, Korea	08/2014
B.S., Industrial and Systems Engineering Georgia Institute of Technology, Atlanta, GA	12/2011

ACADEMIC APPOINTMENTS

Associated Member, Faculty of Bioengineering <i>Texas Tech University</i>	04/2026–Present
Assistant Professor <i>Texas Tech University</i> • Department of Industrial, Manufacturing, and Systems Engineering	09/2025–Present

PUBLICATIONS

Peer-Reviewed Journal Publications (* corresponding-author)

- [J.9] Santos, F., Son, C.*, **Park, H.**, Xue. C. Yang, J., & Liang, N. (2026). Biomechanical effects of virtual reality headset support and task demands on the neck. *IIEE Transactions on Occupational Ergonomics and Human Factors*, 1-19. doi:10.1080/24725838.2026.2680465
- [J.8] Santos, F., Son, C.*, **Park, H.**, Xue. C, Yang, J., & Liang, N. (2026). Design, fabrication, and testing of an ergonomic intervention for a virtual reality headset using 3D printing. *Ergonomics*, 1-15. doi:10.1080/00140139.2026.2653007
- [J.7] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2026). Adaptation to a whole-body powered exoskeleton: Human-exoskeleton coordination during load-handling tasks. *Annals of Biomedical Engineering*, 1-15. doi:10.1007/s10439-026-04025-9
- [J.6] **Park, H.***, & Nussbaum, M.A. (2025). Muscle synergy analysis during pseudo-static and dynamic overhead tasks with arm-support exoskeletons. *Journal of Biomechanics*, 113135. doi:10.1016/j.jbiomech.2025.113135

- [J.5] Ransing, V., Park, J., Ye, Y., **Park, H.**, Kim, S., Du, J., & Srinivasan, D.* (2025). Efficacy of virtual reality-based approach for users to understand the potential benefits and limitations of using exoskeletons. *Human Factors*, 67(11), 1136-1151. doi:10.1177/00187208251346627
- [J.4] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A.* (2025). Passive arm-support and back-support exoskeletons have distinct phase-dependent effects on physical demands during cart pushing and pulling: An exploratory study. *Applied Ergonomics*, 126, 104510. doi:10.1016/j.apergo.2025.104510
- [J.3] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2023). A pilot study investigating motor adaptations when learning to walk with a whole-body powered exoskeleton. *Journal of Electromyography and Kinesiology*, 69, 102755. doi:10.1016/j.jelekin.2023.102755
- [J.2] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2022). Effects of using a whole-body powered exoskeleton during simulated occupational load-handling tasks: A pilot study. *Applied Ergonomics*, 98, 103589. doi:10.1016/j.apergo.2021.103589
- [J.1] **Park, H.**, Park, W.*, & Kim, Y. (2014). Manikin families representing obese airline passengers in the U.S. *Journal of Healthcare Engineering*, 5(4), 479-504. doi:10.1260/2040-2295.5.4.479

Journal Publications in Progress

- [U.1] **Park, H.***, & Nussbaum, M.A. Characterizing adaptation to using exoskeletons: A narrative review and assessment framework for occupational contexts. *In Revision*
- [U.2] **Park, H.***, & Nussbaum, M.A. Perceptual adaptations to back-support exoskeletons: evidence from measures of workload and discomfort. *Submitted*
- [U.3] Santos, F., Son, C.*, **Park, H.**, Xue, C. Yang, J., & Liang, N. Ergonomic Design of Head-Neck Supports Using Additive Manufacturing. *In Revision*
- [U.4] Lee, W., Kwon, S., **Park, H.***, Hwang, D.* When user language meets manufacturing standards: Bridging the gap in 3D printing troubleshooting with domain-specific natural language processing. *In Preparation*
- [U.5] **Park, H.***, & Nussbaum, M.A. Motor adaptations to back-support exoskeletons: evidence from measures of muscle activity, kinematics, and kinetics. *In Preparation*
- [U.6] Jeong, Y.*, Lee, J.H., **Park, H.**, Choi, W.Y., Hwang, D. How Motion Parameters Shape Perceived Usability of an Autonomous Wheelchair: Age-Related Differences Across Deceleration and Speed Modes. *In Preparation*

Peer-Reviewed Conference Proceedings

- [C.6] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *Proceedings of the 66th Human Factors and Ergonomics Society Annual Meeting*, 66(1),

- [C.5] **Park, H.,** Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *Proceedings of the 65th Human Factors and Ergonomics Society Annual Meeting*, 65(1), 1367–1368.
- [C.4] **Park, H.,** Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *Proceedings of the 64th Human Factors and Ergonomics Society Annual Meeting*, 64(1), 888–889. - **Won Best Student Paper Award at the Occupational Ergonomics Technical Group**
- [C.3] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2014). A review on interaction techniques in virtual environments. *Proceedings of the 2014 International Conference on Industrial Engineering and Operations Management* (pp. 1582-1590).
- [C.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.,** Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *Proceedings of the 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*
- [C.1] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *Proceedings of 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

FUNDED RESEARCH PROJECTS AND GRANTS

External Grants

- [EG.1] **Biomechanical and Usability Assessment of Occupational Exoskeletons for Nurses in Overweight and Obese Patient Handling**
 Sponsor: Heartland Center for Occupational Health & Safety Pilot Project Awards;
 sub-award from NIOSH
 Total Amount: \$19,337
 Dates: Jul 2026 – Jun 2027
 Role: Principal Investigator
 Collaborators: Changwon Son

PRESENTATIONS

Conference Oral Presentations

- [O.13] **Park, H.,** & Nussbaum, M.A. (2026). Adaptation to using passive and active back-support exoskeletons during repetitive lifting tasks. *IIE Annual Conference & Expo 2026, Arlington, TX.*
- [O.12] Santos, F., & Son, C., **Park, H.,** (2026). What Happens to Your Neck when You Wear VR for Long: Effects of Task Demands and Ergonomic Supports on Muscle Activity and Kinematics. *IIE Annual Conference & Expo 2026, Arlington, TX.*

- [O.11] Santos, F., & Son, C., **Park, H.**, (2026). What Happens to Your Neck when You Wear VR for Long: Effects of Task Demands and Ergonomic Supports on Muscle Activity and Kinematics. *HFES Regional Meeting, College Station, TX.*
- [O.10] Santos, F., & Son, C., **Park, H.**, (2026). A 3D-printed shoulder support to reduce neck strain during virtual reality device use. *30th Applied Ergonomics Conference, Arlington, TX.*
- [O.9] Santos, F., **Park, H.**, & Son, C. (2025). A 3D-printed shoulder support to reduce neck strain during prolonged virtual reality (VR) device use: pilot findings. *69th Human Factors and Ergonomics Society Annual Meeting, Chicago, IL.*
- [O.8] **Park, H.**, Ojelade A., Kim, S., & Nussbaum, M.A. (2024). Effects of using arm-support exoskeletons on muscle coordination during a pseudo-static overhead task: preliminary analysis of muscle synergy. *68th Human Factors and Ergonomics Society Annual Meeting, Phoenix, AZ. - Won Best Experimental Paper Award at the Occupational Ergonomics Technical Group.*
- [O.7] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A. (2023). Effects of using passive back- and arm-support exoskeletons for cart pushing and pulling. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC.*
- [O.6] Noll, A., Nussbaum, M.A., Kim, S., & **Park, H.** (2023). Reliability Analysis of Task Performance and Subjective Measures for Assessment of Occupational Exoskeletons. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC.*
- [O.5] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *66th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.*
- [O.4] **Park, H.**, Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *65th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.*
- [O.3] **Park, H.**, Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *64th Human Factors and Ergonomics Society Annual Meeting.*
- [O.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.**, Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*
- [O.1] Jung, J., **Park, H.**, Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

Poster Presentations

- [P.3] Santos, F., Son, C., **Park, H.**, Xue. C, Yang, J., & Liang, N. Design, fabrication, and testing of

an ergonomic intervention for virtual reality headset using 3D printing. (2026). *WCOE Research Day, Texas Tech University, Lubbock, TX, February.*

- [P.2] **Park, H.,** Ojelade, A., Kim, S., & Nussbaum, M. A. (2023). Effects of different arm-support exoskeletons on muscle activation patterns during a pseudo-static overhead task - preliminary analysis of muscle synergy. *37th American Society of Safety Professionals (ASSP) Region VI Professional Development Conference (PDC) in Myrtle Beach, South Carolina, September.*
- **Won Student Presentation Award**
- [P.1] **Park, H.,** Kim, S., Nussbaum, M. A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual material handling. *2020 Annual HFES/INFORMS Poster Competition, Virginia Tech, VA, October.* - **Won Student Poster Presentation Award**

Invited Seminars

- [S.3] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Psychological Sciences, Texas Tech University, October.*
- [S.2] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial, Manufacturing, and Systems Engineering, Texas Tech University, April.*
- [S.1] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial and Systems Engineering, Kennesaw State University, March.*

Invited Panel Talk

- [T.1] **Park, H.** (2025). From Student to Professor: Exploring Careers in Occupational Ergonomics. Panel Member. *Occupational Ergonomics Technical Group of HFES, September.*

HONORS & AWARDS

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| • Best Experimental Paper Award, OETG at HFES | 2024 |
| • Best Student Poster Presentation Award, ASSP Regional Conference | 2023 |
| • IISE Future Faculty Fellow (3F) Program Awardee | 2023–2024 |
| • NIOSH Training Fellow, Virginia Tech | 2023–2025 |
| - Fellowship includes monthly stipend, tuition, and required fees | |
| • ISE Graduate Travel Award (\$900), Virginia Tech | 2022 |
| • Best Student Paper Award, OETG at HFES | 2020 |
| • VT HFES Student Chapter Research Competition Award | 2020 |
| • Grado Department of ISE Fellowship, Virginia Tech | 2018 |
| - Fellowship includes monthly stipend, tuition, and required fees | |
| • Brain Korea 21 Scholarship, Seoul National University | 2014 |

- Lecture & Research Scholarship, Seoul National University

2013

TEACHING EXPERIENCE

Instructor of Record

Texas Tech, Dept. of Industrial, Manufacturing, & Systems Engineering

- IE 5304: Biomechanics & Work Physiology (n = 11) Spring, 2026

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 4624: Work Physiology (n = 36) Fall, 2023
- ISE 3614: Human Factors Engineering and Ergonomics (taught online, n = 7) Summer, 2023

Guest Lecturer

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics Spring, 2023
 "NIOSH Lifting Equation and Controls" (n = 100)
- ISE 4644: Occupational Safety and Hazards Control Fall, 2018
 "Review on Hazard Analysis and Prevention" (n = 20)

Graduate Teaching Assistant

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics Spring, 2019
- ISE 4624: Work Physiology Spring, 2019
- ISE 4644: Occupational Health and Hazards Control Fall, 2018

PROFESSIONAL EXPERIENCES

- Software Engineer, Military duty** 10/2014–03/2018
 Kornic Glory, Seoul, Korea
- Research Intern** 06/2011–08/2011
 LG Electronics, Seoul, Korea

ADVISING & MENTORING

Dissertation/Thesis Committee Member

- Felipe Santos, TTU (PhD, IMSE) (2025–Present) [J.9], [J.8], [O.12], [O.11], [O.10], [O.9], [U.2]
 Chair: Changwon Son
- Daniel Leibman, TTU (PhD, Psychology) (2025–Present)
 Chair: Heesun Choi
- Armina Rahman, TTU (PhD, IMSE) (2025–Present)
 Chair: Changwon Son
- Jessica Lillian Dedeaux, TTU (PhD, IMSE) (2026–Present)
 Chair: Changwon Son

Research Mentoring

Voluntary Postdoc Position

- Leonardo Wei, Texas Tech (PhD, ISE) (02/2026–03/2026)

M.S. Students

- Delaney Sheppard, Texas Tech (BIOE) (02/2026–Present)
- Alex Noll, Virginia Tech (ISE) (08/2022–08/2024) [J.4], [C.8], [C.7]

Undergraduate Students

- Enriquez Bazoalto Kaylee, Virginia Tech (ISE) (05/2025-08/2025)

PROFESSIONAL DEVELOPMENT AND SERVICE ACTIVITIES

Professional Development Training

- IISE New Faculty Colloquium - IISE 2026
- IISE Future Faculty Fellow (3F) Program - IISE 2023-2024

Service to Profession

- Session Chair, IISE Annual Conference & Expo 2026, Arlington, TX 2026
- Session Chair, the 69th HFES Annual Meeting, Atlanta, GA 2025

Archival Journal

- Reviewer, *Journal of Biomechanics* 2026–Present
- Reviewer, *Gait and Posture* 2026–Present
- Reviewer, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 2026–Present
- Reviewer, *IEEE Transactions on Human-Machine Systems* 2025–Present
- Reviewer, *IISE Transactions on Occupational Ergonomics and Human Factors* 2023–Present
- Reviewer, *Human Factors and Ergonomics Society Proceedings* 2021–Present

Service to Institution

- Texas Tech Regional Science Fair (K-12) Special Award Judge 2026
- Texas Tech Regional Science Fair (K-12) Ribbon Judge 2026
- Dean's Representative, Dissertation Defense of Yves Valentin (MS in Human Factors Psychology, TTU Department of Psychological Sciences) 2026
- TTU Graduate Academic Advisor 2025–Present
- VT Chapter of the Human Factors and Ergonomics Society (HFES)
 - Officer: *Webmaster* 2020–2022
- National Biomechanics Day
 - *Led lab tours and biomechanics activities for local high school students* 2020–2025
- VT Center for the Enhancement of Engineering Diversity (CEED) TechGirls Summer Camp
 - *Led hands-on biomechanics activities* 2019
- VT College of Engineering Alumni Outreach Event
 - *Led lab tours for alumni* 2019

PROFESSIONAL AFFILIATIONS

Member, Institute of Industrial and Systems Engineers (IISE)	<i>2023–Present</i>
Member, Human Factors and Ergonomics Society (HFES)	<i>2020–Present</i>
Steering Committee member, K-Human Factors and Ergonomics Society (K-HFES)	<i>2021–Present</i>

REMARK

Dual Citizenship: U.S.A. and Republic of Korea